Problem 4.29: Isospin Analysis of $\pi N \to K\Sigma$ Reactions

Griffiths, *Introduction to Elementary Particles*

A detailed, pedagogical walkthrough

Problem Statement

Find the ratio of the cross sections for the following reactions, assuming the CM energy is such that the $I = \frac{3}{2}$ channel dominates:

(a) $\pi^- + p \to K^0 + \Sigma^0$

(b) $\pi^- + p \to K^+ + \Sigma^-$

(c) $\pi^+ + p \to K^+ + \Sigma^+$

What if the energy is such that the $I = \frac{1}{2}$ channel dominates?

The Big Idea: Why Isospin Gives Us Cross-Section Ratios

The strong interaction is (approximately) invariant under rotations in isospin space—it treats u and d quarks identically. Mathematically, the strong Hamiltonian H_s commutes with the isospin operators:

$$[H_s, \mathbf{I}] = 0 \tag{1}$$

The Wigner–Eckart Theorem in Isospin

Because the scattering operator T is an **isoscalar** (rank-0 tensor in isospin space), the Wigner–Eckart theorem tells us:

$$\langle I', I_3' \,|\, T \,|\, I, I_3 \rangle = \delta_{II'}\, \delta_{I_3 I_3'}\, \mathcal{A}_I \tag{2}$$

where $\mathcal{A}_I$ is the **reduced matrix element**—a single complex number that encodes *all* the dynamics of the strong interaction in the channel with total isospin I.

Translation: The T-matrix is *diagonal* in total isospin. Different charge channels of the same underlying reaction share the same reduced amplitudes $\mathcal{A}_I$; they differ only in the Clebsch–Gordan coefficients that project their physical states onto isospin eigenstates.

This means our strategy is:

1. Assign isospin quantum numbers to every particle.
2. Decompose initial and final states into eigenstates of total I.
3. Write each physical amplitude as a sum over isospin channels, weighted by CG coefficients.
4. Take ratios of $|\mathcal{M}|^2$ to get cross-section ratios.

Step 1: Isospin Assignments

Every hadron belongs to an isospin multiplet. The assignments follow from quark content (the strange quark s carries $I = 0$):

Particle	Quarks	I	I_3	S	Multiplet
π^+	$u\bar{d}$	1	$+1$	0	3*Pion triplet
π^0	$(u\bar{u} - d\bar{d})/\sqrt{2}$	1	0	0	
π^-	$d\bar{u}$	1	-1	0	
p	uud	$\frac{1}{2}$	$+\frac{1}{2}$	0	2*Nucleon doublet
n	udd	$\frac{1}{2}$	$-\frac{1}{2}$	0	
K^+	$u\bar{s}$	$\frac{1}{2}$	$+\frac{1}{2}$	$+1$	2*Kaon doublet
K^0	$d\bar{s}$	$\frac{1}{2}$	$-\frac{1}{2}$	$+1$	
Σ^+	uus	1	$+1$	-1	3*Sigma triplet
Σ^0	uds	1	0	-1	
Σ^-	dds	1	-1	-1	

Strangeness Check

All three reactions have $S = 0$ in the initial state and $S = (+1) + (-1) = 0$ in the final state. Strangeness is conserved, confirming these are strong processes where isospin analysis applies. (Weak decays would violate $\Delta S = 0$ and the isospin method would not work.)

Step 2: The Clebsch–Gordan Coefficients for $1 \otimes \frac{1}{2}$

Both the initial state ($\pi \otimes N$: $I = 1$ and $I = \frac{1}{2}$) and the final state ($K \otimes \Sigma$: $I = \frac{1}{2}$ and $I = 1$) involve coupling isospin 1 with isospin $\frac{1}{2}$:

$$1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2} \tag{3}$$

The possible total isospin values are $I = \frac{3}{2}$ and $I = \frac{1}{2}$. Let us write down the CG decompositions we need. We use the notation $|I, I_3\rangle$ for the total isospin states and $|I_1, m_1\rangle|I_2, m_2\rangle$ for the product states.

Building the CG Table from Scratch

Start from the "stretched" state with maximum I_3:

$$\left|\frac{3}{2}, +\frac{3}{2}\right\rangle = |1, +1\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \tag{4}$$

This is the only way to get $I_3 = +\frac{3}{2}$ from combining $m_1 \leq 1$ and $m_2 \leq \frac{1}{2}$.

Apply the lowering operator $I_- = I_{1-} + I_{2-}$ to generate $|3/2, +1/2\rangle$:

$$I_- \left|\frac{3}{2}, +\frac{3}{2}\right\rangle = \sqrt{3}\left|\frac{3}{2}, +\frac{1}{2}\right\rangle \tag{5}$$

$$(I_{1-} + I_{2-})|1, +1\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle = \sqrt{2}\,|1, 0\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle + |1, +1\rangle \left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{6}$$

Dividing by $\sqrt{3}$:

$$\boxed{\left|\frac{3}{2}, +\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\,|1, 0\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}\,|1, +1\rangle \left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \tag{7}$$

The $I = \frac{1}{2}$ state with $I_3 = +\frac{1}{2}$ must be *orthogonal* to the above within the same $I_3 = +\frac{1}{2}$ subspace:

$$\boxed{\left|\frac{1}{2}, +\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\,|1, 0\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\,|1, +1\rangle \left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \tag{8}$$

(The relative sign is fixed by the Condon–Shortley phase convention.)

Applying I_- once more gives the $I_3 = -\frac{1}{2}$ states:

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\,|1, -1\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\,|1, 0\rangle \left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{9}$$

$$\left|\frac{1}{2}, -\frac{1}{2}\right\rangle = -\sqrt{\frac{2}{3}}\,|1, -1\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}\,|1, 0\rangle \left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{10}$$

Quick Reference: The CG Coefficients We Need

The decompositions above can be inverted to express product states in terms of total-I states. The ones we need are:

$$|1, -1\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{11}$$

$$|1, +1\rangle \left|\frac{1}{2}, +\frac{1}{2}\right\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle \tag{12}$$

Coupling Order for the Final State: $\frac{1}{2} \otimes 1$

The final state has K ($I = \frac{1}{2}$) first and Σ ($I = 1$) second, so we need $\frac{1}{2} \otimes 1$ rather than $1 \otimes \frac{1}{2}$. The symmetry relation for CG coefficients gives:

$$\langle j_2, m_2;\, j_1, m_1 \,|\, J, M\rangle = (-1)^{j_1 + j_2 - J}\, \langle j_1, m_1;\, j_2, m_2 \,|\, J, M\rangle \tag{13}$$

With $j_1 = 1$, $j_2 = \frac{1}{2}$:

- For $J = \frac{3}{2}$: phase $= (-1)^{1+1/2-3/2} = (-1)^0 = +1$ (coefficients unchanged)
- For $J = \frac{1}{2}$: phase $= (-1)^{1+1/2-1/2} = (-1)^1 = -1$ (all signs flip)

Using this, the final-state decompositions (in $|I_K, m_K\rangle|I_\Sigma, m_\Sigma\rangle$ notation) are:

$$|K^0\Sigma^0\rangle = \left|\frac{1}{2},-\frac{1}{2}\right\rangle|1,0\rangle = \sqrt{\frac{2}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{14}$$

$$|K^+\Sigma^-\rangle = \left|\frac{1}{2},+\frac{1}{2}\right\rangle|1,-1\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{15}$$

$$|K^+\Sigma^+\rangle = \left|\frac{1}{2},+\frac{1}{2}\right\rangle|1,+1\rangle = \left|\frac{3}{2},+\frac{3}{2}\right\rangle \tag{16}$$

Watch the Signs!

Comparing (14) with Griffiths' solution: Griffiths writes the $K^0\Sigma^0$ final state with a $+\sqrt{1/3}$ coefficient for $|1/2,-1/2\rangle$, whereas the $\pi^- p$ initial state has $-\sqrt{2/3}$ for the same isospin state. These relative signs are crucial—they determine whether $\mathcal{A}_{3/2}$ and $\mathcal{A}_{1/2}$ add or subtract in each amplitude. The sign difference comes from the coupling-order symmetry relation above.

Step 3: Computing the Physical Amplitudes

The physical amplitude for each reaction is:

$$\mathcal{M} = \sum_I \langle \text{final} \,|\, I, I_3\rangle \; \mathcal{A}_I \; \langle I, I_3 \,|\, \text{initial}\rangle \tag{17}$$

where the sum runs over $I = \frac{3}{2}$ and $I = \frac{1}{2}$, and I_3 is fixed by charge conservation.

Reaction (a): $\pi^- p \to K^0\Sigma^0$ $\quad (I_3 = -\frac{1}{2})$

Initial state from (11):

$$|\pi^- p\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{18}$$

Final state from (14):

$$|K^0\Sigma^0\rangle = \sqrt{\frac{2}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{19}$$

Amplitude: We pick out the coefficient of each $|I, I_3\rangle$ in both states:

$$\mathcal{M}_a = \underbrace{\sqrt{\frac{2}{3}}}_{\substack{\langle K^0\Sigma^0| \\ I=3/2}} \mathcal{A}_{3/2} \underbrace{\sqrt{\frac{1}{3}}}_{\substack{\langle 3/2| \\ \pi^- p\rangle}} + \underbrace{\sqrt{\frac{1}{3}}}_{\substack{\langle K^0\Sigma^0| \\ I=1/2}} \mathcal{A}_{1/2} \underbrace{\left(-\sqrt{\frac{2}{3}}\right)}_{\substack{\langle 1/2| \\ \pi^- p\rangle}}$$

$$= \frac{\sqrt{2}}{3}\mathcal{A}_{3/2} - \frac{\sqrt{2}}{3}\mathcal{A}_{1/2} \tag{20}$$

$$\boxed{\mathcal{M}_a = \frac{\sqrt{2}}{3}\left(\mathcal{A}_{3/2} - \mathcal{A}_{1/2}\right)} \tag{21}$$

Wait—this differs from Griffiths' answer!

Griffiths obtains $\mathcal{M}_a = \frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2})$. The difference is a *sign convention* in how the $K\Sigma$ final state CG coefficients are defined (coupling order $\frac{1}{2} \otimes 1$ vs. $1 \otimes \frac{1}{2}$). Both conventions give the **same cross-section ratios** because the sign of $\mathcal{A}_{1/2}$ relative to $\mathcal{A}_{3/2}$ is not physically observable—only $|\mathcal{M}|^2$ matters. When we take ratios in the limiting cases ($\mathcal{A}_{1/2} \to 0$ or $\mathcal{A}_{3/2} \to 0$), the sign is irrelevant. We can therefore proceed with *either* convention and get the same final answer.

To match Griffiths exactly, let us adopt his convention and write:

$$\mathcal{M}_a = \frac{\sqrt{2}}{3}\left(\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\right) \qquad \text{(Griffiths convention)}$$

Reaction (b): $\pi^- p \to K^+\Sigma^-$ $\quad (I_3 = -\frac{1}{2})$

Initial state: same as (a).

Final state from (15):

$$|K^+\Sigma^-\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{22}$$

Amplitude:

$$\begin{aligned}\mathcal{M}_b &= \sqrt{\frac{1}{3}}\,\mathcal{A}_{3/2}\,\sqrt{\frac{1}{3}} + \left(-\sqrt{\frac{2}{3}}\right)\mathcal{A}_{1/2}\,\left(-\sqrt{\frac{2}{3}}\right) \\ &= \frac{1}{3}\,\mathcal{A}_{3/2} + \frac{2}{3}\,\mathcal{A}_{1/2}\end{aligned} \tag{23}$$

In Griffiths' convention:

$$\boxed{\mathcal{M}_b = \frac{1}{3}\,\mathcal{A}_{3/2} - \frac{2}{3}\,\mathcal{A}_{1/2}} \qquad \text{(Griffiths convention)}$$

(Again, the sign of the $\mathcal{A}_{1/2}$ term depends on convention; what matters is that it has twice the magnitude of the $\mathcal{A}_{3/2}$ coefficient, with opposite sign.)

Reaction (c): $\pi^+ p \to K^+\Sigma^+$ $\quad (I_3 = +\frac{3}{2})$

Initial state from (12):

$$|\pi^+ p\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle \tag{24}$$

Final state from (16):

$$|K^+\Sigma^+\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle \tag{25}$$

Why This Channel is Pure $I = 3/2$

Both $\pi^+ p$ and $K^+\Sigma^+$ have $I_3 = +\frac{3}{2}$. Since the maximum I_3 from $1 \otimes \frac{1}{2}$ is $+\frac{3}{2}$, and only $I = \frac{3}{2}$ can have $I_3 = +\frac{3}{2}$, these states are *pure* $I = \frac{3}{2}$. There is simply no room for any $I = \frac{1}{2}$ component. This is the "stretched state" argument.

Amplitude:

$$\boxed{\mathcal{M}_c = \mathcal{A}_{3/2}} \tag{26}$$

This is the simplest of the three: both CG coefficients are 1.

Step 4: Cross-Section Ratios

Since the cross section is proportional to $|\mathcal{M}|^2$ (all kinematic factors being the same for reactions at the same CM energy with the same initial state—or related by isospin for different initial states at the same $\sqrt{s}$):

$$\sigma_a \propto \frac{2}{9}\left|\mathcal{A}_{3/2}+\mathcal{A}_{1/2}\right|^2, \qquad \sigma_b \propto \frac{1}{9}\left|\mathcal{A}_{3/2}-2\mathcal{A}_{1/2}\right|^2, \qquad \sigma_c \propto \left|\mathcal{A}_{3/2}\right|^2 \tag{27}$$

Case 1: $I = 3/2$ Dominates ($\mathcal{A}_{1/2} \approx 0$)

Setting $\mathcal{A}_{1/2} = 0$:

$$\sigma_a : \sigma_b : \sigma_c = \frac{2}{9}|\mathcal{A}_{3/2}|^2 : \frac{1}{9}|\mathcal{A}_{3/2}|^2 : |\mathcal{A}_{3/2}|^2 \tag{28}$$

Multiplying through by 9:

$$\boxed{\sigma_a : \sigma_b : \sigma_c = 2 : 1 : 9} \tag{29}$$

Physical Interpretation

The $\pi^+ p$ channel dominates overwhelmingly because it feeds *entirely* into $I = 3/2$ with unit coefficient. The $\pi^- p$ channels only access a fraction of the $I = 3/2$ amplitude through their CG projections ($\sqrt{1/3}$ for the initial state). This situation arises physically near a Δ-like resonance with $I = 3/2$.

Case 2: $I = 1/2$ Dominates ($\mathcal{A}_{3/2} \approx 0$)

Setting $\mathcal{A}_{3/2} = 0$:

$$\sigma_a : \sigma_b : \sigma_c = \frac{2}{9}|\mathcal{A}_{1/2}|^2 : \frac{4}{9}|\mathcal{A}_{1/2}|^2 : 0 \tag{30}$$

Multiplying through by $9/(2|\mathcal{A}_{1/2}|^2)$:

$$\boxed{\sigma_a : \sigma_b : \sigma_c = 1 : 2 : 0} \tag{31}$$

A Striking Prediction: $\sigma_c = 0$

The reaction $\pi^+ p \to K^+ \Sigma^+$ is **completely forbidden** when only $I = 1/2$ contributes! This is because $|\pi^+ p\rangle$ is a stretched state with $I_3 = +3/2$—it is *pure* $I = 3/2$ and has **zero overlap** with the $I = 1/2$ amplitude.

Experimental consequence: If you observe $\sigma_c \neq 0$ at some energy, there *must* be an $I = 3/2$ contribution. Conversely, at energies dominated by N^* resonances with $I = 1/2$, this channel should nearly vanish.

Summary of Results

Reaction	**Amplitude $\mathcal{M}$**	$\lvert\mathcal{M}\rvert^2$
(a) $\pi^- p \to K^0\Sigma^0$	$\frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2})$	$\frac{2}{9}\lvert\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\rvert^2$
(b) $\pi^- p \to K^+\Sigma^-$	$\frac{1}{3}(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2})$	$\frac{1}{9}\lvert\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\rvert^2$
(c) $\pi^+ p \to K^+\Sigma^+$	$\mathcal{A}_{3/2}$	$\lvert\mathcal{A}_{3/2}\rvert^2$

Regime	$\boldsymbol{\sigma_a : \sigma_b : \sigma_c}$
$I = 3/2$ dominates	$2 : 1 : 9$
$I = 1/2$ dominates	$1 : 2 : 0$

Consistency Check: An Amplitude Relation

As a nice sanity check, note that the three amplitudes satisfy:

$$\mathcal{M}_c = \sqrt{2}\,\mathcal{M}_a + \mathcal{M}_b \tag{32}$$

Proof:

$$\begin{aligned}\sqrt{2}\,\mathcal{M}_a + \mathcal{M}_b &= \sqrt{2}\cdot\frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2}) + \frac{1}{3}(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}) \\ &= \frac{2}{3}\mathcal{A}_{3/2} + \frac{2}{3}\mathcal{A}_{1/2} + \frac{1}{3}\mathcal{A}_{3/2} - \frac{2}{3}\mathcal{A}_{1/2} \\ &= \mathcal{A}_{3/2} = \mathcal{M}_c \quad \checkmark\end{aligned} \tag{33}$$

This identity holds for *any* values of $\mathcal{A}_{3/2}$ and $\mathcal{A}_{1/2}$—it is a model-independent consequence of isospin symmetry, testable by measuring three cross sections and their relative phases. Such amplitude triangle relations are a hallmark of the isospin formalism.

The Deeper Lesson: Geometry vs. Dynamics

This problem beautifully illustrates the separation of **geometry** (group theory, CG coefficients) from **dynamics** (the reduced matrix elements $\mathcal{A}_I$, which encode all the difficult physics of confinement, resonances, and final-state interactions).

You never need to compute $\mathcal{A}_I$ from first principles to predict the cross-section ratios in the limiting cases. When one isospin channel dominates—as happens near a resonance—the ratios become *pure numbers* determined entirely by CG coefficients.

The analogy with ordinary angular momentum is exact: just as selection rules for electromagnetic transitions follow from the Wigner–Eckart theorem applied to spatial rotations, these cross-section relations follow from applying the *same theorem* to rotations in isospin space.